

Comparing Schedule-Based Low-Rank Functionals with KSS, BLM, and Borovičková-Shimer

Purpose and reading guide

This note asks a simple simulation question: when the true worker-firm wage schedule is known, how well do several procedures recover the population objects they are meant to estimate?

Two terms are important. An **estimand** is the true population quantity we want to learn. An **estimator** is the rule that uses a finite sample to estimate it. Different procedures do not always have the same estimand. In particular, KSS is built around an additive worker-firm model, while the project's estimands use the complete wage schedule and can include worker-firm interactions. The simulation therefore separates:

- 1 estimation error relative to each procedure's own, or *native*, estimand;
- 2 the gap between that native estimand and the project's estimand.

The project procedure currently implemented is the **low-rank plug-in without LOO correction**. The full leave-worker-out estimator is still under development, so this note does not claim to evaluate it.

The experiment has seven designs and 100 replications per design. Because the simulator creates the complete population wage schedule, every population target can be calculated exactly before sampling begins.

Statements labeled **Source** report a definition, method, design choice, or stored numerical result. Statements labeled **Derived** follow from those definitions or summarize the stored Monte Carlo output.

Source: SIMULATION_CONTRACT.md; configs/full_ladder.json; PRODUCTION_RESULTS.md; reporting commit 8922da6; production configuration fingerprint e03e0c242706502fcd58f20c1cc52c4deeb234f48c981a4885e13157d6ac2a09.

Notation

Source: project notation and estimand definitions in SIMULATION_CONTRACT.md; BLM and BS notation is specialized from the cited papers as described in Sections 2.3 and 2.4 below.

Symbol	Meaning	Why it matters
i, j	Worker and firm indices; i, j denote random workers and firms.	They identify a cell in the worker-by-firm wage schedule.
Y_{ij}	Observed wage, or log wage, for worker i at firm j .	This is what the estimator sees in sampled matches.
m_{ij}	Systematic wage for worker i at firm j , before wage noise.	The simulator knows every m_{ij} , including unobserved matches.
$M = (m_{ij})$	Complete systematic wage schedule.	Project estimands are functionals of this matrix.
p^{obs}	Population assignment probabilities over worker-firm pairs.	These probabilities describe who works where.
p_i, q_j	Worker and firm marginals of p^{obs} .	They weight workers and firms in the project estimands.
μ, a_i, b_j, h_{ij}	Grand mean, worker main effect, firm main effect, and interaction in the schedule decomposition.	They separate additive wage variation from worker-firm-specific gains.
Q_F	Average firm-side wage variation across workers.	It measures how much a worker's systematic wage changes across firms.
H_F	Average variation across workers in pairwise firm wage differences.	It measures heterogeneity in workers' relative rankings of firms.
ρ_H	Interaction share $H_F/(2Q_F)$.	It records the share of firm-side wage variation due to interactions.
C_{assign}	Assignment contribution to systematic-wage variance.	It compares observed assignment with random matching that preserves p_i and q_j .
V_{ψ}^{AKM}	Firm-effect variance in the additive AKM projection.	Under nonadditivity, it need not equal Q_F .
A_{lk}^{BLM}	Mean wage for worker type l and firm class k .	This is the grouped BLM mean target.
ρ_{BS}	Correlation between BS worker and firm wage types on observed matches.	It is a sorting measure, but it is not C_{assign} .
$\hat{r}_{BIC}$	Rank selected by the exploratory common-support BIC rule.	It chooses among the fitted low-rank models.
e_b	Estimate minus target in replication b .	Bias and RMSE are computed from these errors.

1. Model and population targets

1.1 What the simulator generates

The project model is

$$Y_{ij} = X'_{ij}\delta + \alpha_i + \psi_j + u'_i\Lambda v_j + \varepsilon_{ij}.$$

The additive terms α_i and ψ_j give worker and firm main effects. The term $u'_i\Lambda v_j$ allows the value of a firm to differ across workers. This is the nonadditive interaction. The final term ε_{ij} is wage noise.

The current simulation sets $X'_{ij}\delta = 0$. It first creates the systematic part of the wage for every possible worker-firm pair:

$$m_{ij} = \alpha_i + \psi_j + u'_i\Lambda v_j.$$

It then draws actual matches using p^{obs} and adds wage noise. Thus the estimators see only a noisy, incomplete panel, while the researcher can use the full M to calculate ground truth.

Source: project model and SIMULATION_CONTRACT.md, Sections 1 and 4.

1.2 Separating main effects from interactions

To define the project estimands, imagine randomly pairing workers and firms using the observed worker shares p_i and firm shares q_j . Under this product distribution, the complete schedule can be written as

$$m_{ij} = \mu + a_i + b_j + h_{ij}.$$

Here, μ is the overall mean. The term a_i is worker i 's average wage advantage, b_j is firm j 's average wage advantage, and h_{ij} is what remains for that particular worker-firm pair. The weighting rules make each remaining component average to zero along the relevant dimension, so the decomposition is unique.

This decomposition gives an intuitive distinction:

- **Additive separability:** $h_{ij} = 0$ for every pair. Workers can have different wage levels, but they agree on firm wage differences.
- **Nonadditive separability:** some $h_{ij} \neq 0$. A firm can be especially valuable to one worker and less valuable to another.

Source: SIMULATION_CONTRACT.md, Sections 1 and 2.

1.3 The four project estimands

The first estimand is

$$Q_F = \sum_i p_i \text{Var}_q(m_{ij}).$$

For each worker, Q_F calculates wage variation across firms and then averages that variance across workers. It includes both average firm differences and worker-firm interactions.

The second estimand is

$$H_F = \sum_{j,k} q_j q_k \text{Var}_p(m_{ij} - m_{ik}).$$

For each pair of firms, H_F asks whether their wage difference is the same for all workers. If every worker values firms j and k by the same amount, the variance for that pair is zero. Averaging across firm pairs makes H_F a measure of interaction heterogeneity.

The normalized interaction share is

$$\rho_H = \frac{H_F}{2Q_F}.$$

The factor of two comes from the identity

$$Q_F = \text{Var}_q(b_j) + \frac{1}{2}H_F.$$

Thus, when $Q_F > 0$, ρ_H is the fraction of firm-side wage variation associated with worker-firm interactions rather than the common firm component.

Finally, the assignment estimand is

$$C_{\text{assign}} = \frac{1}{2}[\text{Var}_{p^{\text{obs}}}(m_{ij}) - \text{Var}_{p \times q}(m_{ij})].$$

The first variance uses observed assignment: workers are matched to firms as in the population. The second randomly pairs them while preserving worker and firm frequencies. A positive C_{assign} means observed assignment creates more systematic-wage dispersion than this random-matching reference.

Under additive separability, the definitions simplify to

$$H_F = 0, \quad \rho_H = 0, \quad Q_F = \text{Var}_q(b_j), \quad C_{\text{assign}} = \text{Cov}_{p^{\text{obs}}}(a_i, b_j).$$

This special case matters because the project targets then coincide with the usual additive firm-variance and worker-firm covariance targets.

Source: SIMULATION_CONTRACT.md, Section 2. Derived explanations follow by substituting the product-weighted decomposition into the four definitions.

1.4 Why procedures can disagree even when both are well estimated

Under nonadditivity, an additive estimator and a schedule-based estimator can be accurate for different population quantities. For any procedure p ,

$$\hat{\theta}^{(p)} - \theta_{\text{project}} = (\hat{\theta}^{(p)} - \theta_p^*) + (\theta_p^* - \theta_{\text{project}}).$$

The first term is **estimation error**: did procedure p recover its own native target θ_p^* ? The second is the **estimand gap**: is that native target the same object as the project target? Reporting both prevents a difference in scientific questions from being mistaken for a bad estimator.

Derived algebraic decomposition from the declared targets in SIMULATION_CONTRACT.md, Section 5.

2. Procedures compared

2.1 Project low-rank plug-in without LOO correction

The observed panel contains repeated wages for some worker-firm pairs and no wage for many other pairs. Let $\bar{Y}_{ij}$ be the mean observed wage and n_{ij} the number of observations for an observed pair. For a chosen rank r , the current project procedure solves

$$\min_{\alpha, \psi, U, V} \sum_{(i,j) \in E} n_{ij} (\bar{Y}_{ij} - \alpha_i - \psi_j - u_i' v_j)^2,$$

where E is the set of observed worker-firm pairs. The fitted low-rank model then fills the missing cells of the wage schedule. In this objective, Λ is absorbed into the fitted factors U and V . This is why the interaction is written as $u_i' v_j$ rather than $u_i' \Lambda v_j$; it is a different parameterization of the same low-rank interaction matrix. The four project functionals are calculated from the completed schedule.

The implementation uses weighted alternating least squares, several starting values, factor centering, and singular-value normalization. Positive-rank models are fitted on a connected support core with enough distinct matches per worker and firm. An exploratory BIC rule compares ranks on the same observed sample.

The procedure also labels a fit *unstable* when its numerical diagnostics indicate unreliable completion. Its finite output is still included in unconditional performance because a user applying the procedure would receive that value. Stable-only summaries are reported separately to diagnose what happens when completion works.

No quadratic or leave-worker-out correction is applied. Therefore this is a plug-in benchmark, not the project's proposed full LOO estimator.

Source: SIMULATION_CONTRACT.md, Sections 4 and 7; src/loo_sim/low_rank.py; src/loo_sim/monte_carlo.py.

2.2 AKM and KSS

AKM fits an additive worker and firm fixed-effects model. Its plug-in moments use the estimated firm effects and worker effects directly. KSS-HO adjusts quadratic moments under homoskedastic errors, while KSS-HE uses a heteroskedastic leave-out correction. Kline, Saggio, and Sølvesten (2020) develop leave-out estimators for quadratic forms in linear models.

When the true schedule is nonadditive, the native population target of AKM/KSS in this experiment is the best additive approximation under observed assignment:

$$(\mu^{\text{AKM}}, \alpha^{\text{AKM}}, \psi^{\text{AKM}}) = \arg \min_{\mu, \alpha, \psi} \sum_{i,j} P_{ij}^{\text{obs}} (m_{ij} - \mu - \alpha_i - \psi_j)^2.$$

The comparison moments are

$$V_{\psi}^{\text{AKM}} = \text{Var}_q(\psi_j^{\text{AKM}})$$

and the corresponding assignment-weighted covariance between worker and firm effects. Under additive separability, these targets equal Q_F and C_{assign} . Under nonadditivity, the additive projection leaves out the interaction, so the native targets can differ from the project targets. The simulation therefore reports both native-target bias and the native-to-project estimand gap.

Source: Kline, Saggio, and Sølvesten (2020), *Econometrica* 88(5), 1859-1898, DOI 10.3982/ECTA16410; PyTwoWay 0.3.21; src/loo_sim/pytwo_way_estimators.py; SIMULATION_CONTRACT.md, Section 5.1.

2.3 Grouped BLM

BLM allows wages to depend nonlinearly on a worker's latent type and a firm's latent class. Its grouped mean target is

$$A_{\ell k}^{\text{BLM}} = \mathbb{E}[Y \mid L = \ell, K = k].$$

The simulation considers two versions. The oracle version is given the true firm classes. The estimated-group version clusters firms using their empirical wage distributions. Estimated labels are aligned to the simulated labels only when calculating Monte Carlo error.

This procedure should be especially well suited to the grouped design because that design is built from discrete worker types and firm classes. The comparison tests whether exploiting that discrete structure performs better than fitting continuous individual-level low-rank factors.

Source: Bonhomme, Lamadon, and Manresa (2019), *Econometrica* 87(3), 699-739, DOI 10.3982/ECTA15722; local stationary specialization in SIMULATION_CONTRACT.md, Section 5.2, and src/loo_sim/blm.py.

2.4 Borovičková-Shimer

The Borovičková-Shimer procedure forms worker wage types and firm wage types from observed matches and estimates their correlation. Its population analogue in the simulation is

$$\rho_{\text{BS}} = \text{Corr}_{P^{\text{obs}}}(\lambda_i^{\text{BS}}, \mu_j^{\text{BS}}).$$

This is a correlation-based measure of assortative matching. It is not renamed as C_{assign} , which is a variance difference for the complete schedule. The simulation evaluates the estimator against ρ_{BS} , its own target.

Source: Borovičková and Shimer (2020), "High Wage Workers Work for High Wage Firms," February 2020 manuscript; earlier version NBER Working Paper 24074, DOI 10.3386/w24074; local target in SIMULATION_CONTRACT.md, Section 3.

3. Monte Carlo design

The seven designs form a ladder. Each early step changes one feature so that a change in performance has a clear interpretation. The final two designs then test discrete grouping and higher rank.

Design	Population schedule	Assignment	Observations per worker	Candidate ranks
Additive, independent	80 workers, 10 firms, rank 0	Independent	$T = 7$	0, 1
Additive, common sorting	80 workers, 10 firms, rank 0	Common sorting 0.8	$T = 7$	0, 1
Rank 1, independent	80 workers, 10 firms, one continuous interaction factor	Independent	$T = 7$	0, 1
Rank 1, common sorting	80 workers, 10 firms, one continuous interaction factor	Common sorting 0.8	$T = 7$	0, 1
Rank 1, interaction sorting	80 workers, 10 firms, one continuous interaction factor	Interaction sorting 0.4	$T = 10$	0, 1
Grouped BLM	300 workers, 18 firms; 2 worker types by 3 firm classes	Common sorting 0.4; interaction sorting 0.2	$T = 5$	0, 1
Rank 2	80 workers, 10 firms; interaction singular values 1 and 0.5	Common sorting 0.4; interaction sorting 0.4	$T = 15$	0, 1, 2

The first design is the easiest benchmark: the model is additive and matching is independent. The second adds sorting while keeping the wage model additive. Designs three through five introduce a rank-one interaction and then change how workers select firms. The grouped design asks whether BLM's discrete structure helps when it matches the data-generating process. The last design checks whether the low-rank procedure can distinguish rank two from rank zero or one.

All designs use a wage-noise standard deviation of 0.5. The probability of redrawing a match is 0.75, except in the grouped design, where it is 0.35. Each design has 100 replications. The production run contains 32,400 scalar estimate-target records and 5,200 estimator attempts: 4,773 successful, 427 unstable, and no hard failures.

For replication error e_b , the reported statistics are

$$\text{Bias} = \frac{1}{B} \sum_{b=1}^B e_b,$$

$$\text{MCSE}(\text{Bias}) = \frac{\text{sd}(e_b)}{\sqrt{B}},$$

$$\text{RMSE} = \sqrt{\frac{1}{B} \sum_{b=1}^B e_b^2}.$$

Bias shows the average direction of error. MCSE describes simulation uncertainty in the reported bias. RMSE combines bias and dispersion, so it penalizes both systematic error and rare large errors.

Source: configs/full_ladder.json; PRODUCTION_RESULTS.md; src/loo_sim/monte_carlo.py. The three performance formulas are calculated from stored replication-level errors.

4. Results

4.1 Additive designs: a common-target comparison

Additive designs: common targets, different finite-sample behavior

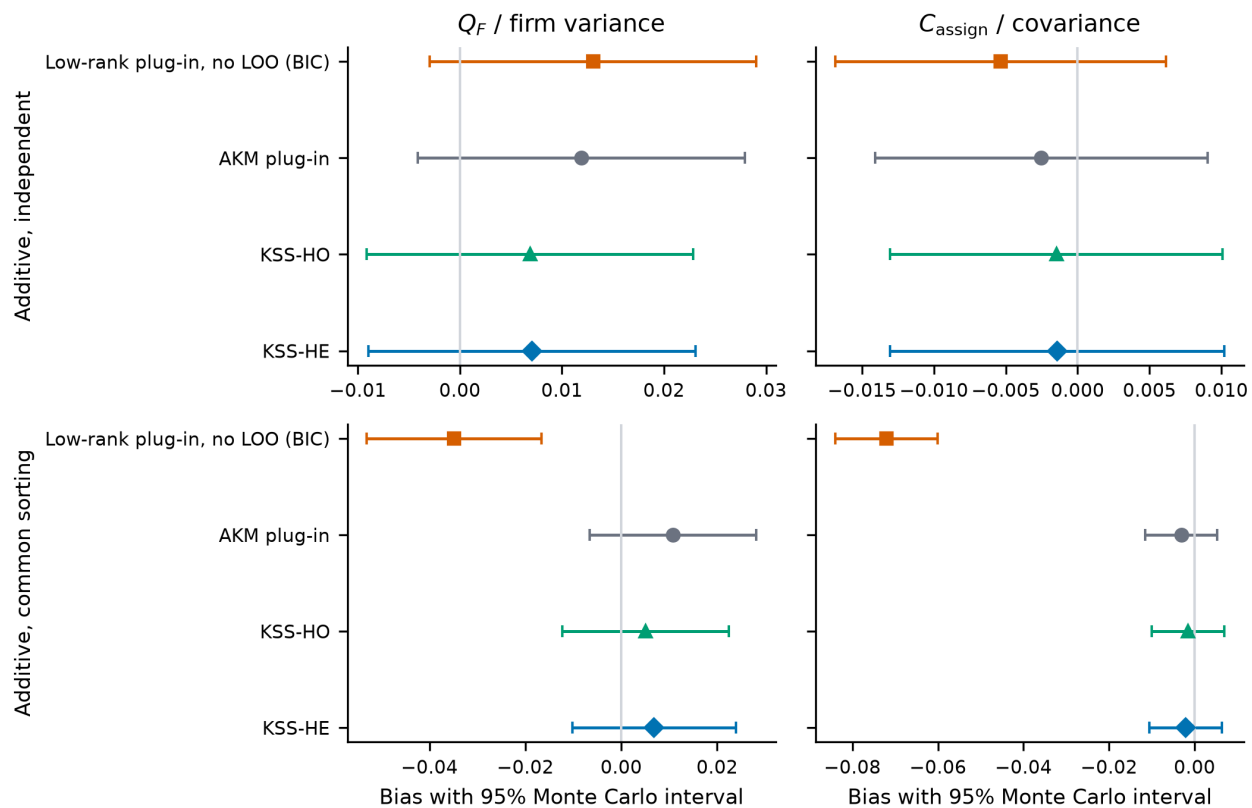


Figure 1. Additive-design bias and 95 percent Monte Carlo intervals.

The additive designs provide the cleanest comparison because the project, AKM, and KSS firm-variance and assignment targets coincide.

With independent assignment, the low-rank plug-in, AKM plug-in, KSS-HO, and KSS-HE have nearly identical RMSE. The low-rank plug-in bias is 0.013 for Q_F and -0.005 for C_{assign} . KSS-HE bias is 0.007 and -0.001, respectively.

With common sorting, the targets still coincide, but the low-rank plug-in bias becomes -0.035 with MCSE 0.009 for Q_F and -0.072 with MCSE 0.006 for C_{assign} . KSS-HE bias remains small: 0.007 with MCSE 0.009 and -0.002 with MCSE 0.004.

Because the population targets are equal here, the difference under sorting is a finite-sample estimator difference, not an estimand gap. The unfinished LOO correction is intended to address plug-in bias, but these results cannot show how that future estimator will perform.

Source: reports/production/tables/additive_comparison.csv. Derived interpretation uses the additive identities in Section 1.3.

4.2 Selecting the correct rank does not guarantee stable completion

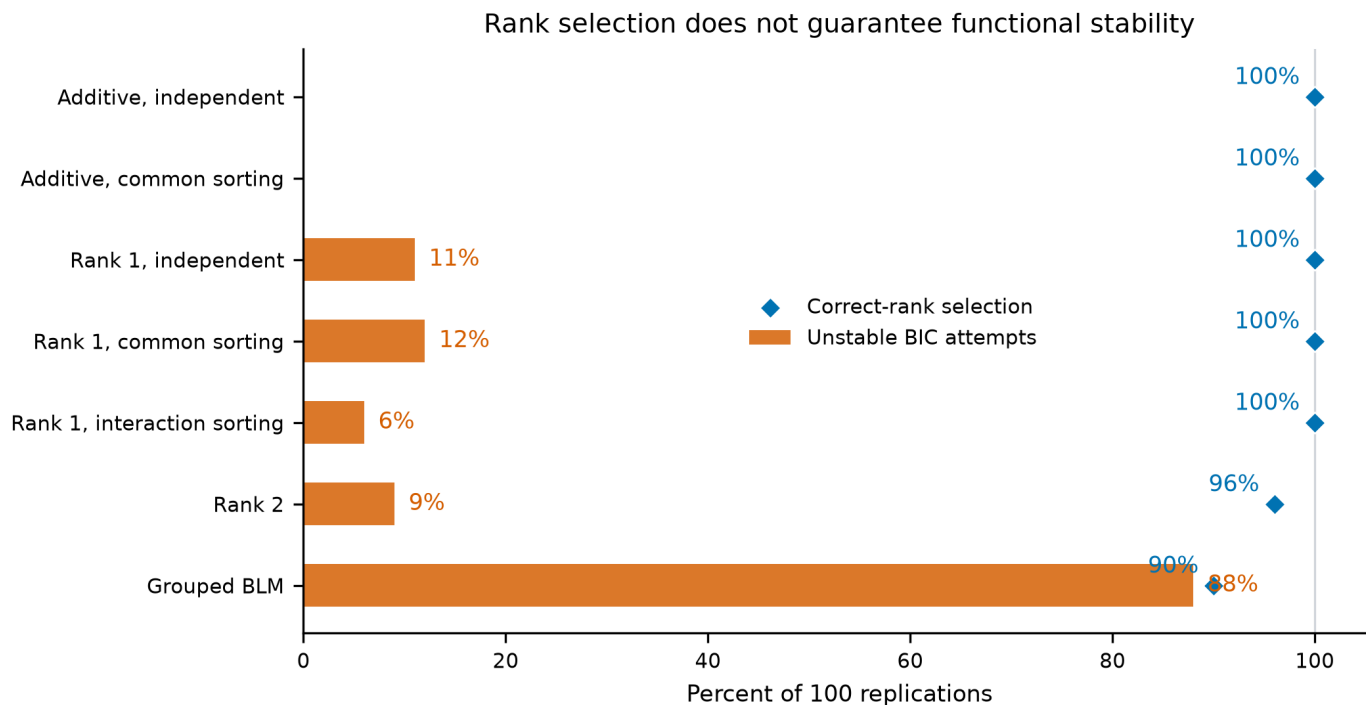


Figure 2. BIC rank selection and unstable-attempt frequencies.

BIC selects the true rank in every additive and rank-one continuous-factor replication and in 96 percent of rank-two replications. However, 6 to 12 percent of the selected rank-one and rank-two fits are classified as unstable. In the grouped design, BIC selects rank one in 90 percent of replications, but 88 percent of those fits are unstable.

These facts answer two different questions. Rank selection asks which candidate model best fits the observed cells. Stability asks whether that fit can safely fill the many unobserved cells needed for the project functionals. A model can fit observed cells well and still extrapolate poorly.

Source: reports/production/tables/stability_rank_selection.csv and stored attempt diagnostics. The distinction between fit and completion is derived from the estimator construction in Section 2.1.

4.3 Nonadditivity: stable-only accuracy versus actual procedure risk

Nonadditive low-rank plug-in (no LOO): instability drives tail risk

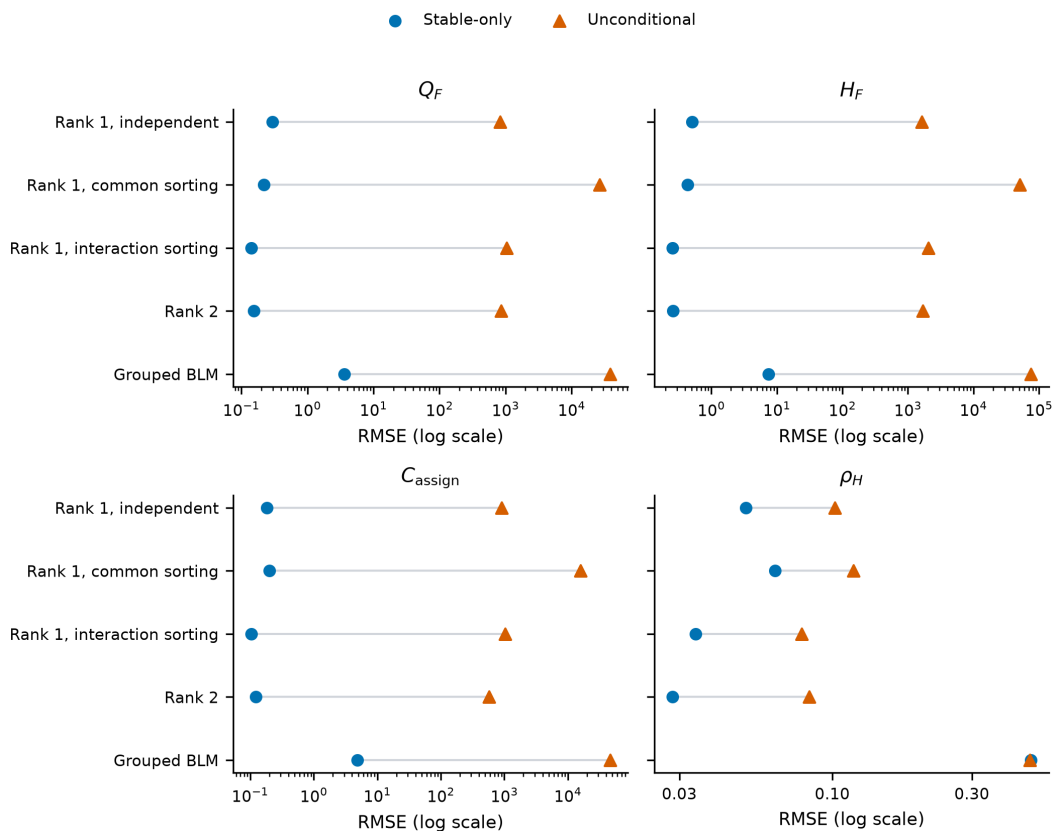


Figure 3. Unconditional and stable-only RMSE for project functionals.

Among stable continuous-factor fits, RMSE ranges from 0.142 to 0.296 for Q_F , 0.255 to 0.513 for H_F , 0.104 to 0.201 for C_{assign} , and 0.028 to 0.064 for ρ_H . Thus, when the completion is stable, the plug-in can recover the four schedule-based objects with moderate error.

The unconditional result is much worse. The 6 to 12 percent unstable continuous-factor fits sometimes produce extremely large functionals. As a result, unconditional Q_F RMSE ranges from 823 to 26,563. In the grouped design, only 12 of 100 low-rank fits are stable. Stable-only Q_F RMSE is already 3,586, and unconditional Q_F RMSE is about 38,100.

Stable-only results reveal the estimator's behavior after a fit-dependent screen. They are useful for diagnosing the source of failure, but they do not describe what happens when the procedure is run without knowing the answer in advance. The unconditional result is therefore the main measure of current operational risk.

Source: reports/production/tables/nonadditive_comparison.csv; PRODUCTION_RESULTS.md, nonadditive-results section. The interpretation follows from the reporting rule in SIMULATION_CONTRACT.md, Section 7.

4.4 KSS can estimate its native target without estimating Q_F

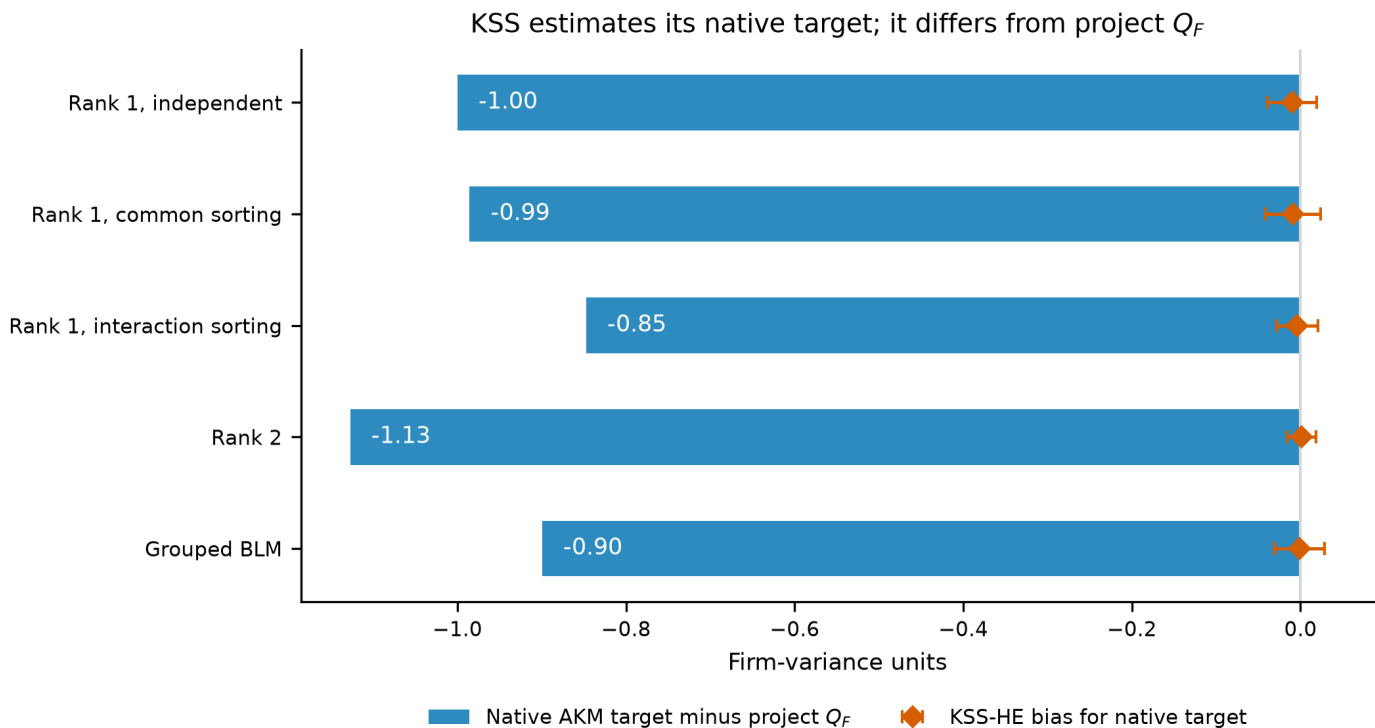


Figure 4. Native AKM/KSS target gaps and KSS-HE native-target bias.

Across the five nonadditive designs, the native AKM firm-variance target is 0.847 to 1.127 below the project target Q_F . In contrast, KSS-HE bias relative to that native AKM target ranges from -0.010 to 0.001, with MCSE between 0.009 and 0.017.

The result follows the two-part comparison in Section 1.4. KSS-HE has small estimation error for the additive projection, but the additive projection itself omits much of the wage variation captured by Q_F . This is not a failure of KSS. It shows that a precisely estimated additive firm variance can answer a different question from the project's schedule-based firm functional. KSS also does not produce H_F or ρ_H .

Source: reports/production/tables/nonadditive_comparison.csv; native-target definitions in SIMULATION_CONTRACT.md, Section 5.1. The two-part reading is derived from the decomposition in Section 1.4.

4.5 The grouped design favors grouped BLM

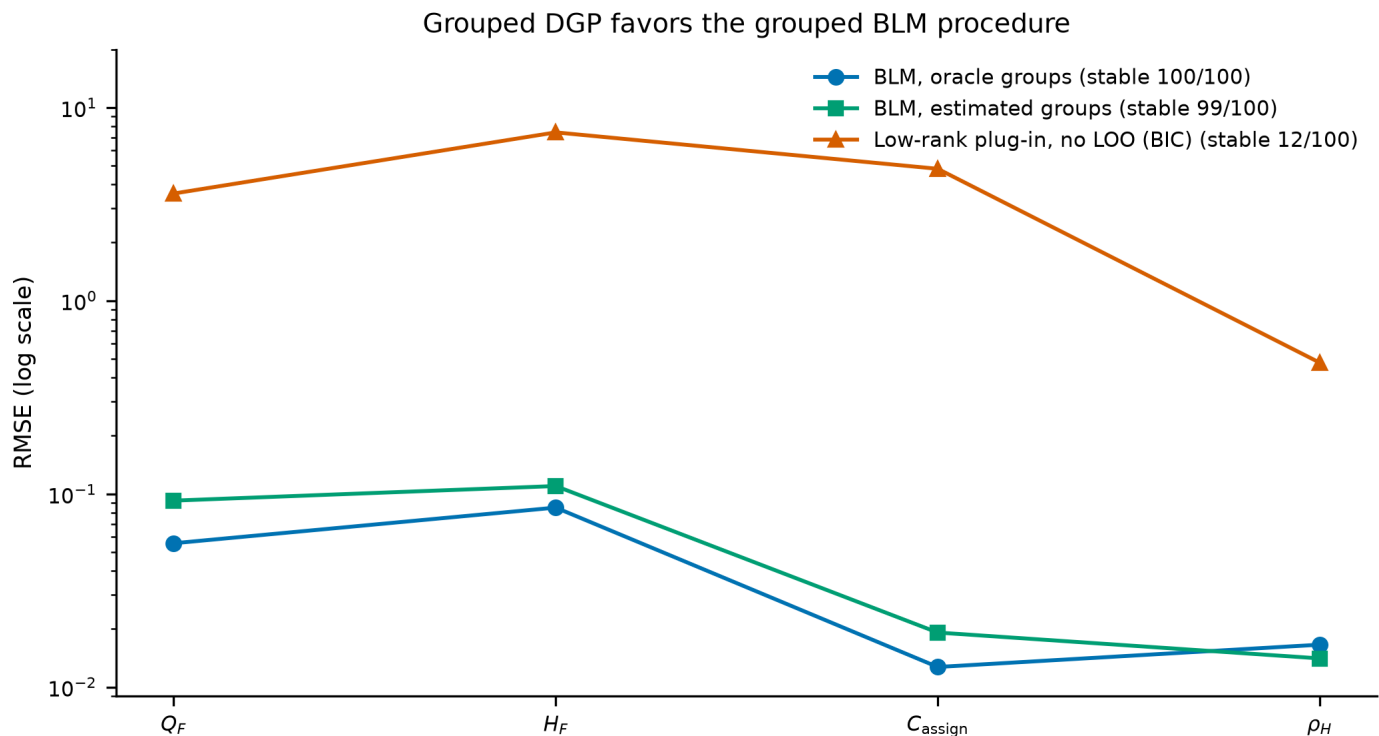


Figure 5. Stable-only RMSE in the grouped BLM design.

Oracle-group BLM is stable in all 100 replications, and estimated-group BLM is stable in 99. Both have aligned cell-mean RMSE of about 0.051.

For Q_F , H_F , C_{assign} , and ρ_H , oracle-group RMSE is 0.056, 0.085, 0.013, and 0.017. Estimated-group RMSE is 0.092, 0.110, 0.019, and 0.014. The individual-level low-rank plug-in is stable in only 12 replications; even within those 12, its RMSE is 3.586, 7.431, 4.814, and 0.478.

BLM performs better here because its discrete worker-type by firm-class structure matches the way the grouped data are generated. Low matrix rank alone is not enough to ensure reliable individual-level completion when the observed worker-firm network is sparse.

Source: reports/production/tables/grouped_blm_comparison.csv. The final explanation is derived from the grouped DGP and the reported stability and RMSE values.

4.6 Borovičková-Shimer estimates a different sorting object



Figure 6. Bias of the Borovičková-Shimer native worker-firm correlation estimator.

The implemented Borovičková-Shimer procedure returns a value in every replication, but it is negatively biased for its native correlation target. Bias ranges from -0.049 in the grouped design to -0.404 in the rank-two design. In the additive-independent design, the bias is -0.149. The procedure also uses a smaller valid sample after removing return spells and workers or firms with too few observations.

These are finite-sample results for the implemented cleaning and estimation pipeline. They do not imply that ρ_{BS} and C_{assign} are interchangeable. The first is a correlation between worker and firm wage types on observed matches; the second is a variance difference calculated from the complete schedule.

Source: reports/production/tables/bs20_native.csv; SIMULATION_CONTRACT.md, Section 3; PyTwoWay 0.3.21 wrapper in src/loo_sim/pytwoWay_estimators.py.

5. What the experiment establishes

The results support three conclusions, in this order.

- 1 **When targets coincide, estimator differences are visible.** In the additive designs, the project, AKM, and KSS targets are the same. KSS-HE remains nearly unbiased under sorting, while the current low-rank plug-in shows finite-sample bias.
- 2 **Under nonadditivity, the target must be stated before comparing estimates.** KSS-HE estimates its additive-projection target accurately, but that target is well below Q_F . BLM performs well when the DGP has the grouped structure it is designed to use. The BS estimator concerns a separate correlation target.
- 3 **The current low-rank plug-in has a completion problem, not only a rank selection problem.** Stable fits can have moderate error, but a small share of unstable fits creates extremely large unconditional RMSE. Correct rank selection does not remove this tail risk.

The experiment does **not** evaluate standard errors for the unfinished project estimator, show that the eventual LOO correction removes the observed bias, or prove that the current stability rule is optimal. The natural next experiment is to keep this DGP ladder fixed and replace the plug-in with the completed leave-worker-out estimator once its correction and feasible standard-error formula are finalized.

Derived synthesis from Figures 1-6 and their source tables. Scope limitation: SIMULATION_CONTRACT.md, estimator-label and reporting contracts.

References

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- Project simulation sources: `SIMULATION_CONTRACT.md`, `configs/full_ladder.json`, `PRODUCTION_RESULTS.md`, and `reports/production/report_metadata.json`.